

Trigonometric functions

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Trigonometric Ratios For an angle.

For any acute angle θ we define trigonometric ratios with reference to a right angled triangle containing θ .

These are Sine (sin), cosine (cos), Tangent (tan), cosecant (cosec), Secant (sec) and Cotangent (cot).

The ratios are defined as follows.

$$(i) \sin \theta = \frac{PQ}{OQ}$$

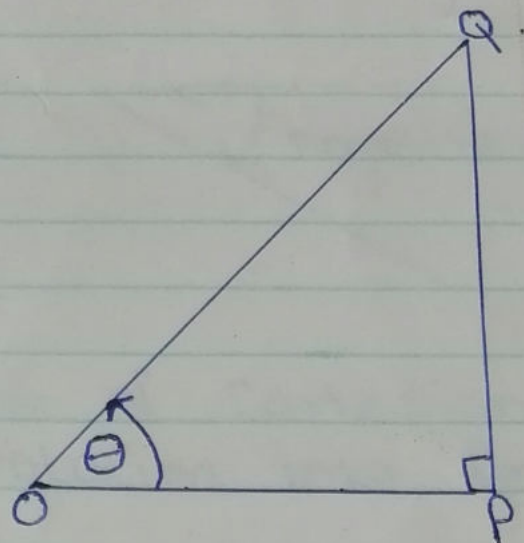
$$(ii) \cos \theta = \frac{OP}{OQ}$$

$$(iii) \tan \theta = \frac{PQ}{OP}$$

$$(iv) \operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{OQ}{PQ}$$

$$(v) \sec \theta = \frac{1}{\cos \theta} = \frac{OQ}{OP}$$

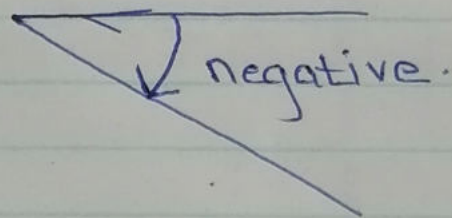
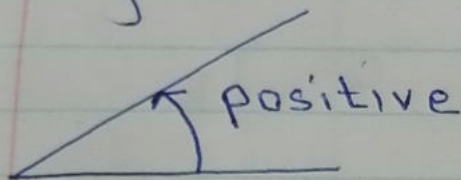
$$(vi) \cot \theta = \frac{1}{\tan \theta} = \frac{OP}{PQ}$$



How about angles that are not acute?
NB: An angle is a measure of rotation from a given position of straight line about a fixed point.

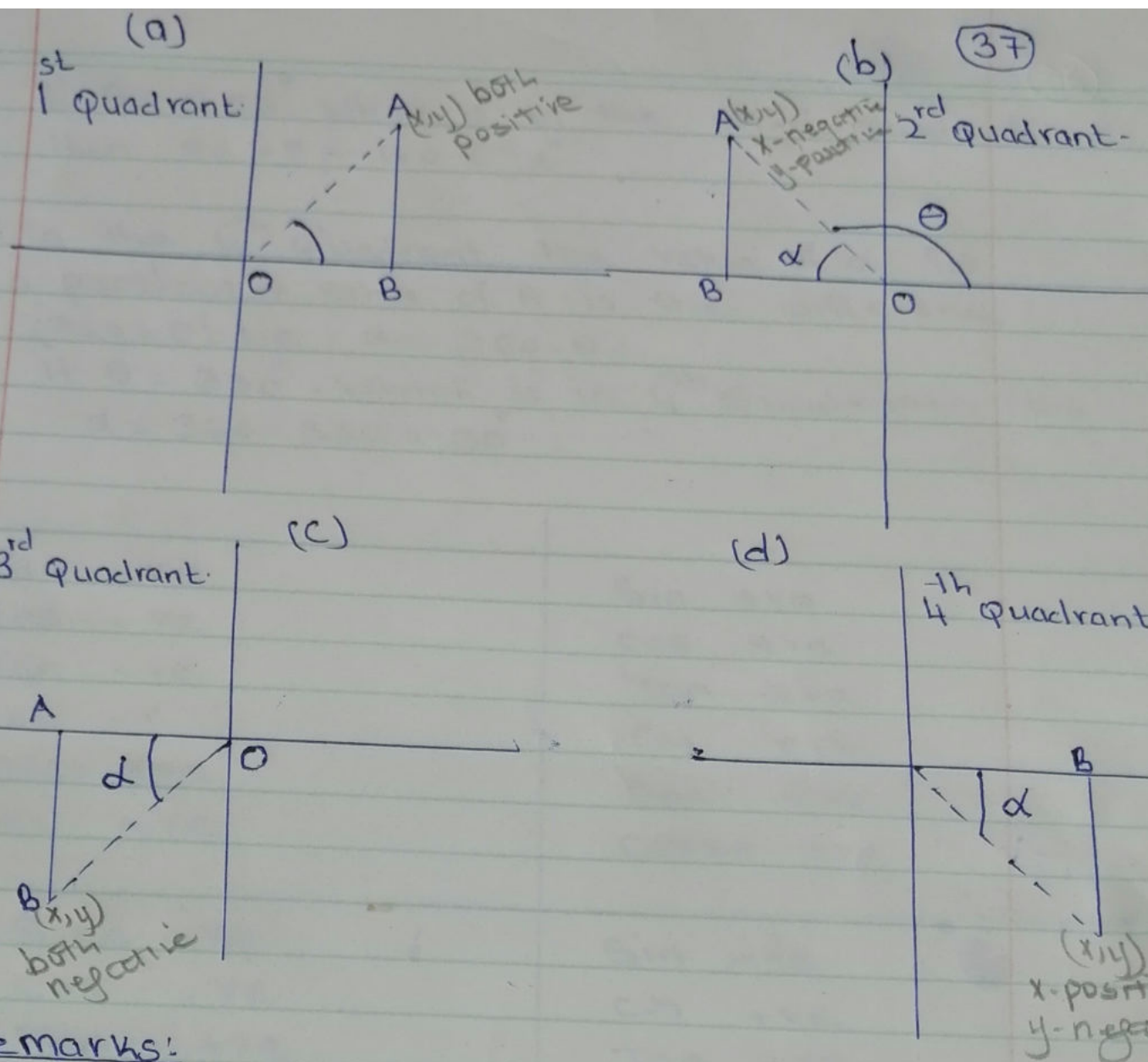
Using this then a general angle is measured about the point of rotation of a line from an initial line which is x-axis on x-y plane.

An angle formed when the line rotates anticlockwise is taken as positive while clockwise rotation corresponds to negative angles.



We know that the pair of Cartesian divides the plane of rotation into four quadrants.

Now if we form a right angled triangle in any of the four quadrants we obtain an acute angle, Say α , regardless of the angle Say θ that Op makes with x-axis



Remarks:

(a) in 1st Quadrant, $\alpha = \theta$

(b) in 2nd Quadrant, the value of α for a particular value of θ is the difference between θ and $\pi (180^\circ)$ i.e. $\alpha = 180^\circ - \theta$

e.g. if $\theta = 140^\circ$ which is in 2nd quadrant,
 $\alpha = 180 - 140 = 40^\circ$

(c) in 3rd Quadrant, the value of α for a particular value of θ is the difference $(\theta - 180^\circ)$
 i.e. $\alpha = \theta - 180^\circ$

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If $\theta = 252^\circ$ which is in the third quadrant,
then $\alpha = 252 - 180 = 72^\circ$

(d) In the 4th quadrant, the value of α for a particular value of θ is the difference $(360 - \theta)$ i.e. $\alpha = 360 - \theta$.

If $\theta = 330^\circ$, which is in 4th quadrant, then
 $\alpha = 360 - 330 = 30^\circ$.

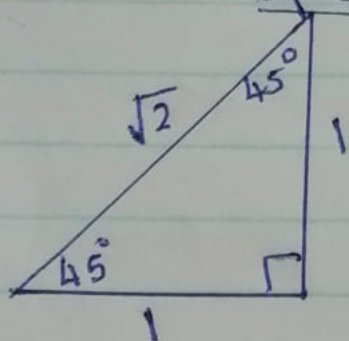
Sin +ve
Cos -ve
Tan -ve
Cot -ve
Cosec +ve
Sec -ve

Sin +ve
Cos +ve
Tan +ve
Cot +ve
Sec +ve
Cosec +ve

Sine -ve
Cos -ve
Tan +ve
Cot +ve
Cosec -ve
Sec -ve

Sin -ve
Cos +ve
Tan -ve
Cosec -ve
Sec +ve
Cot -ve

Special angles:



$$\Rightarrow \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

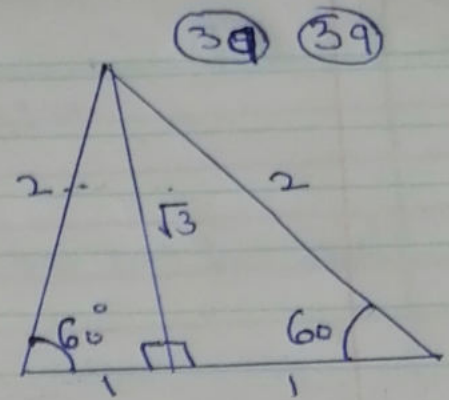
$$\tan 45^\circ = 1.$$

$$\Rightarrow \cos 30^\circ = \frac{\sqrt{3}}{2} = \sin 60^\circ$$

$$\sin 30^\circ = \frac{1}{2} = \cos 60^\circ$$

Since the two angles are complementary angles

$$\tan 30^\circ = \frac{1}{\sqrt{3}} \text{ and } \tan 60^\circ = \sqrt{3}$$



Summary of angles

Angle	30°	45°	60°
Sine	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
Cosine	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
Tangent	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$
Cotangent	$\sqrt{3}$	1	$1/\sqrt{3}$
Cosecant	2	$\sqrt{2}$	$2/\sqrt{3}$
Secant	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2

Exercise:

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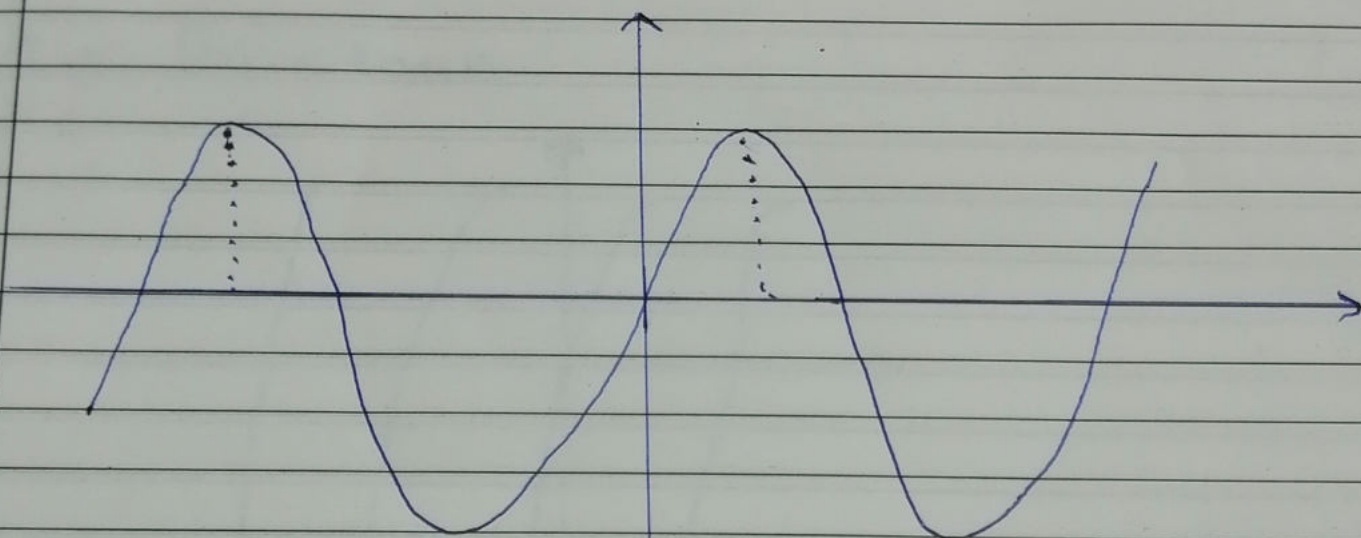
Without using tables or a calculator evaluate the following

(i) $\cos 120^\circ$ (ii) $\tan 150^\circ$ (iii) $\cot 135^\circ$ (iv) $\sin 240^\circ$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

Since there is a value of each trigonometric ratio for any angle, we are now able to plot a graph showing the behaviour of each trigonometric function as the value of θ varies

For Example, let $f(\theta) = \sin \theta$



This graph is continuous

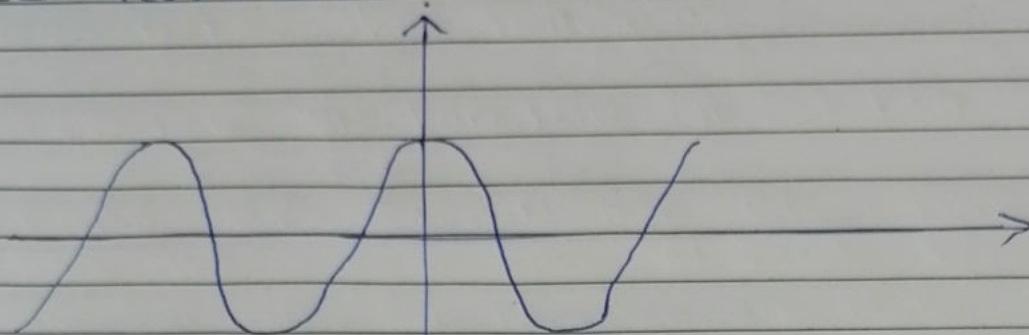
It lies entirely in the range $-1 \leq \sin \theta \leq 1$.

The sine function is an odd function

It is a periodic function of 2π .

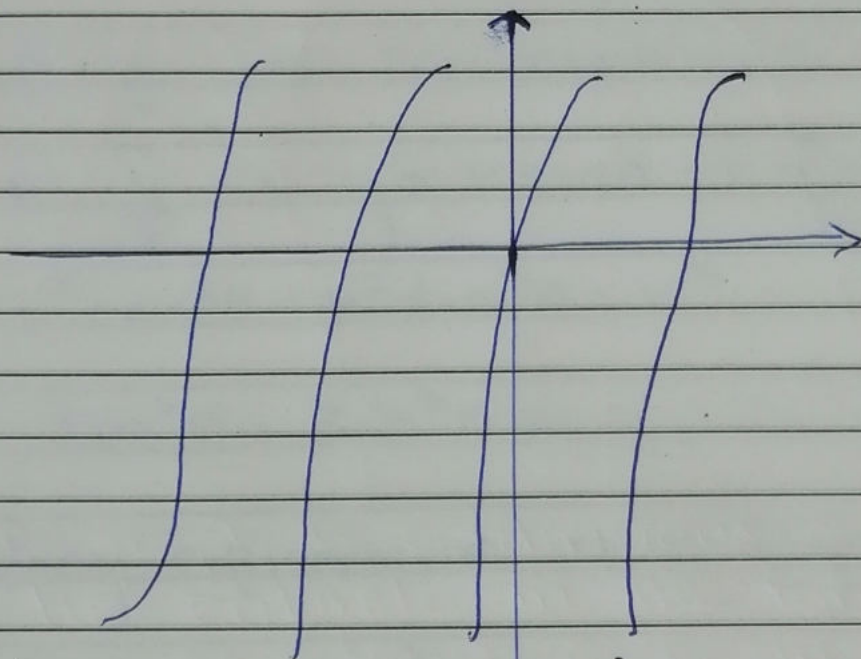
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Let $f(\theta) = \cos \theta$.



- Again the $\cos$ function is continuous
- It lies entirely in the range $-1 \leq \cos \theta \leq 1$.
- Cosine function is an even function.

Let $f(\theta) = \tan \theta$.



This graph is not continuous.
The range is unlimited.
It is a periodic function with period of π .

Exercise:

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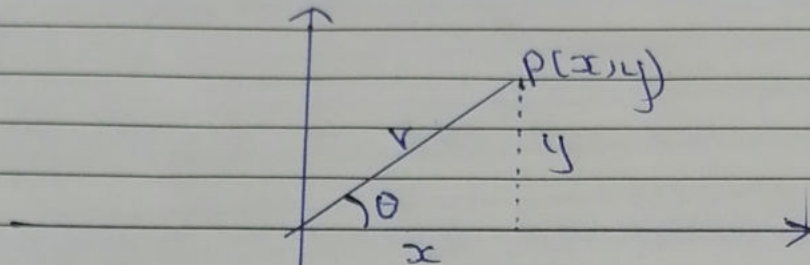
Sketch the graphs of the following functions

(i) $f(x) = \sin \frac{x}{2}$

(ii) $f(x) = \cos 3x$

(iii) $f(x) = \tan 4x$

TRIGONOMETRIC IDENTITIES



We know that $x^2 + y^2 = r^2$ (Pythagoras theorem)

$$\cos \theta = \frac{x}{r} \Rightarrow x = r \cos \theta \Rightarrow x^2 = r^2 \cos^2 \theta$$

$$\sin \theta = \frac{y}{r} \Rightarrow y = r \sin \theta \Rightarrow y^2 = r^2 \sin^2 \theta$$

$$\Rightarrow r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta = 1 \dots \dots \dots (1)$$

Dividing equation all through by $\sin^2 \theta$

$$\Rightarrow \frac{\cos^2 \theta}{\sin^2 \theta} + \frac{\sin^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$$

$$\text{but } \frac{\cos \theta}{\sin \theta} = \cot \theta \Rightarrow \frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta$$

$$\frac{1}{\sin \theta} = \operatorname{cosec} \theta \Rightarrow \frac{1}{\sin^2 \theta} = \operatorname{cosec}^2 \theta$$

$$\Rightarrow \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta \dots \dots \dots (2)$$

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Again dividing equation (1) all through with $\cos^2 \theta$ we obtain

$$\frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta \cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$\text{but } \frac{\sin \theta}{\cos \theta} = \tan \theta \Rightarrow \frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$$

$$\frac{1}{\cos^2 \theta} = \sec^2 \theta \Rightarrow \frac{1}{\cos^2 \theta} = \sec^2 \theta$$

$$\Rightarrow \tan^2 \theta + 1 = \sec^2 \theta \dots \dots \dots (3)$$

Equations (1), (2) and (3) are known as Pythagorean identities.

Addition Formulae.

- (i) $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$
- (ii) $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
- (iii) $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
- (iv) $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$

The above four equations are referred to as Compound angle identities or Addition formulae.

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$$\text{Since } \tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

For Example.

$$\begin{aligned}\cos 105^\circ &= \cos(60^\circ + 45^\circ) \\ &= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ\end{aligned}$$

Double Angle formulae.

By using the compound angle formulae above, we introduce another set of formulae.

Recall that

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\text{If } \beta = \alpha \Rightarrow$$

$$(i) \cos(\alpha + \alpha) = \cos^2 \alpha - \sin^2 \alpha$$

$$\Rightarrow \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$(ii) \sin(\alpha + \alpha) = \sin \alpha \cos \alpha + \cos \alpha \sin \alpha$$

$$\Rightarrow \sin(2\alpha) = 2 \sin \alpha \cos \alpha$$

$$(iii) \tan(\alpha + \alpha) = \frac{\tan \alpha + \tan \alpha}{1 - \tan \alpha \tan \alpha} = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

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But we know that
 $\sin^2 \alpha + \cos^2 \alpha = 1$.

$$\begin{aligned}\Rightarrow \cos(2\alpha) &= \cos^2 \alpha - \sin^2 \alpha \\ &= (1 - \sin^2 \alpha) - \sin^2 \alpha \\ &= 1 - 2\sin^2 \alpha.\end{aligned}$$

$$\begin{aligned}\text{or } \cos(2\alpha) &= \cos^2 \alpha - (1 - \cos^2 \alpha) \\ &= \cos^2 \alpha - 1 + \cos^2 \alpha \\ &= 2\cos^2 \alpha - 1.\end{aligned}$$

Half angle Identity

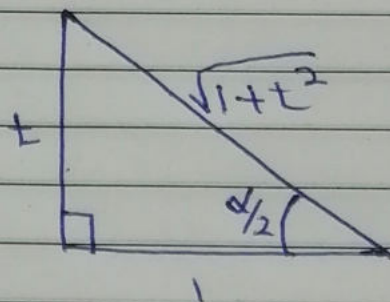
From the double angle formulae

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}.$$

If we let $2\alpha = \theta \Rightarrow \tan \alpha = \tan \frac{\theta}{2}$.

$$\text{let } \tan \frac{\theta}{2} = t$$

$$\Rightarrow \cos \frac{\theta}{2} = \frac{1}{\sqrt{1+t^2}}$$



$$\sin \frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}}.$$

$$\text{But } \cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\Rightarrow \cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$$

$$= \frac{1}{1+t^2} - \frac{t^2}{1+t^2} = \frac{1-t^2}{1+t^2}.$$

$$\Rightarrow \cos \theta = \frac{1-t^2}{1+t^2}.$$

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$$\text{Again } \sin 2\theta = 2 \sin \theta \cos \theta$$
$$\Rightarrow \sin \theta = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2}$$

$$\Rightarrow \sin \theta = \frac{2 \times \frac{t}{\sqrt{1+t^2}} \times \frac{1}{\sqrt{1+t^2}}}{1+t^2} = \frac{2t}{1+t^2}$$

Exponential and Logarithmic Functions (47)

Exponential and logarithmic functions are inverse of each other.

Exponential Function

For a fixed value of the base a the expression a^x clearly varies with the value of x . e.g. $(0.5)^x$, 1^x , 2^x , 3^x and so on.

$\Rightarrow$ It is a function of x .

The functions a^x (as different values are chosen for a) are called exponential functions.

The most and widely used exponential function has the particular base

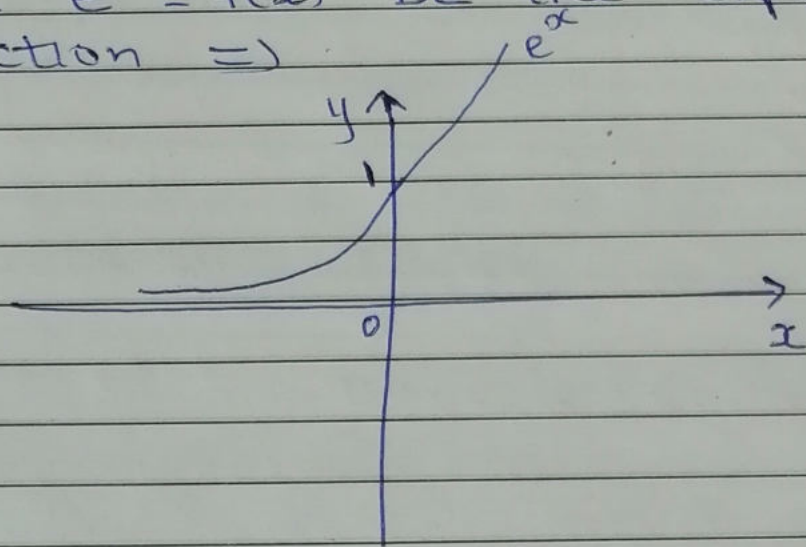
$$e = 2.7182818 \dots$$

Recall that e is an irrational number $\Rightarrow$

It cannot be expressed as a fraction

The approximate value of e is stored in most calculators.

Let $e^x = f(x)$ be the exponential function $\Rightarrow$



Exponential functions appear in various areas of mathematics and engineering.

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For example, the shape of a hanging chain or rope, under the effect of gravity, is well described by a combination of the exponential curves e^{kx} , e^{-kx} .

The function e^{-x^2} plays a major role in Statistics; It being fundamental in the important normal distribution which describes the variability in many naturally occurring phenomena.

The exponential function e^{-kx} appears again in the area of Statistics, in the poisson distribution which (amongst other things) is used to predict the number of events which occur randomly in a given time interval.

The expression e^x as $x \rightarrow \infty$ is called the exponential growth.

The expression e^{-x} as $x \rightarrow \infty$ is called the exponential decay.

The domain of the exponential function is $\mathbb{R}$ (the Set of real numbers) and the range is the set of nonnegative real numbers.

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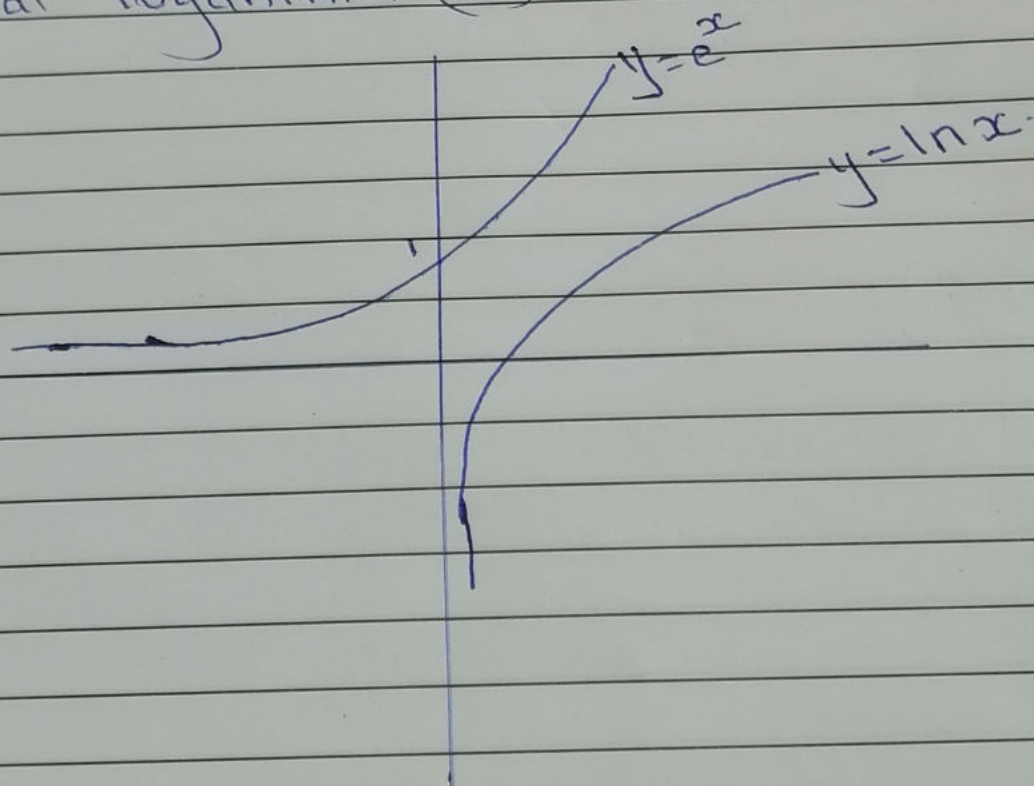
The natural exponential function is the inverse of the natural logarithm

$$\text{i.e. } y = \exp x \text{ iff } x = \ln y.$$

$$\Rightarrow y = e^x \text{ iff } x = \ln y.$$

$$\text{NB} = \ln y = \ln e^y$$

Natural logarithm ($\ln$) is to base e .



Theorem (i) $\ln e^x = x$ for every x

(ii) $e^{\ln x} = x$ for every $x > 0$

The domain is $(0, +\infty)$ and range is $(-\infty, \infty)$